

Technology Stack

Date	28 September 2026
Team ID	SWTID-2026-6259
Project Name	AI Study Buddy API
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js	Provides the frontend interface for user registration, login, study material upload, summaries, quizzes, flashcards, and study plans.

2	Backend / Server-Side	Node.js + Express.js	Used to build the backend RESTful APIs, handle requests, authentication, routing, AI services, and database operations.
3	Database / Data Storage	MongoDB + Mongoose	MongoDB stores users, study materials, and AI-generated resources. Mongoose provides data modeling, schema validation, and database interaction.
4	Cloud / Hosting / Deployment	Not specified in the demo	The LearnMate demo describes local execution using Node.js and MongoDB but does not specify a cloud hosting or deployment platform.
5	Version Control & CI/CD	Not specified in the demo	The demo does not specify a version-control or CI/CD technology.

6	Third-Party APIs / Other Tools	Google Gemini 2.5 Flash API	Generates AI-powered summaries, flashcards, multiple-choice quizzes, and personalized study plans from study materials.
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